



Project Management Skills-20PM01T _ Notes REV-01-2024

Unit 05: Project Control, Review and Audit**Topics to be covered:**

- 5.1 Introduction to Project control
- 5.2 Project Control Purposes
- 5.3 Problems of Project Control
- 5.4 Gantt Charts 5.5 Milestone Charts
- 5.6 Critical Path Method (CPM)
- 5.7 Construction of a Network
- 5.8 Network Technique in Project Scheduling (PERT)
- 5.9 Project Review 5.10 Initial Review and its types
- 5.11 Objectives of Project Audit
- 5.12 Functions of Project Auditor
- 5.13 Case Studies

5.1 Introduction to Project control

- ☐ **Project control** is the process of monitoring and managing a project's scope, schedule, cost and quality to complete the project successfully.
- ☐ **Project control** involves collecting data, analyzing and taking corrective actions.

5.2 Project Control Purposes**Following are the purposes of Project Control:**

- To complete the project as per the plan.
- To identify the problems and solve them.
- To control the risk through risk management.
- To manage resources and reduce wastages.
- To control the project time.
- To control the project cost.
- To control the project scope.
- To control project quality.
- To control delays in the project.
- To control the project team and process through monitoring.
- To follows proper standards and to complete the project effectively and efficiently.

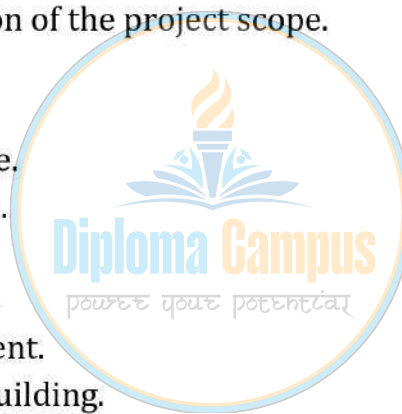
Importance of Project Control

- ❑ Project control is important for the project manager to track project progress as per the plan.
 - Project control prevents cost overruns.
 - Project control prevents time overruns.
 - Project control make sure project objectives are completed efficiently.
 - Project control helps in decision-making through real-time data analysis.
 - Project control improves project success rates by reducing project risks.

5.3 Problems of Project Control

Following are the Problems in Project Control:

- Unplanned expansion of the project scope.
- Time overrun
- Cost overrun.
- Political interference.
- Poor administration.
- Poor planning.
- Poor project design.
- Poor risk management.
- Poor project team building.
- Wrong choice of equipment.
- Incompetent material suppliers.
- Inaccurate initial cost estimation.
- Ineffective project time management.
- Failures in project performance.
- A change in the scope of the project.
- Delays in starting and executing some of the project activities.
- A delay in one project, results in delays in subsequent projects.
- Use of out-dated technology.



5.4 Gantt Charts

- ☐ A Gantt Chart is a project management tool used to plan, schedule and track tasks in a project timeline.
- ☐ It provides a visual representation of tasks to be completed in a project.
- ☐ It shows project task duration, its dependencies and progress.
- ☐ It shows all the activity that need to be completed.
- ☐ It shows when activity it will start, how long it may take and when can it be finish.
- ☐ Gantt Chart are like a roadmap which project team can refer to complete project activities on given time and given cost.

Structure of a Gantt Chart are as follows:

A Gantt chart consists of:

- **Horizontal Bars:** Each bar represents a task and its duration.
- **Timeline (X-Axis):** Displays dates or time periods.
- **Tasks (Y-Axis):** Lists all project activities.

How Gantt Charts Work

- **Tasks are listed** on the left side.
- **Time** (days, weeks or months) are shown at the top.
- **Bars** represent each task's start and end date.
- **Progress tracking** is possible by updating the chart as the project moves forward. (if new tasks are added it can be continued in the bottom)

Benefits of Using Gantt Charts

- **Better Planning:** Clearly defines task sequences and deadlines.
- **Improved Time Management:** Helps track progress and delays.
- **Resource Allocation or Resource management:** Assigns tasks efficiently.
- **Find Task Dependencies:** Shows which tasks rely on others.
- **Improve Communication:** Provides a shared project overview for teams.

Q. Develop the Gantt chart for the given project: (Aug-Sep-2022)

JOBS	START DAY	DURATION	MAN POWER
J-1	0	5	7
J-2	2	3	3
J-3	4	6	9
J-4	8	4	2
J-5	11	4	4

Solution:

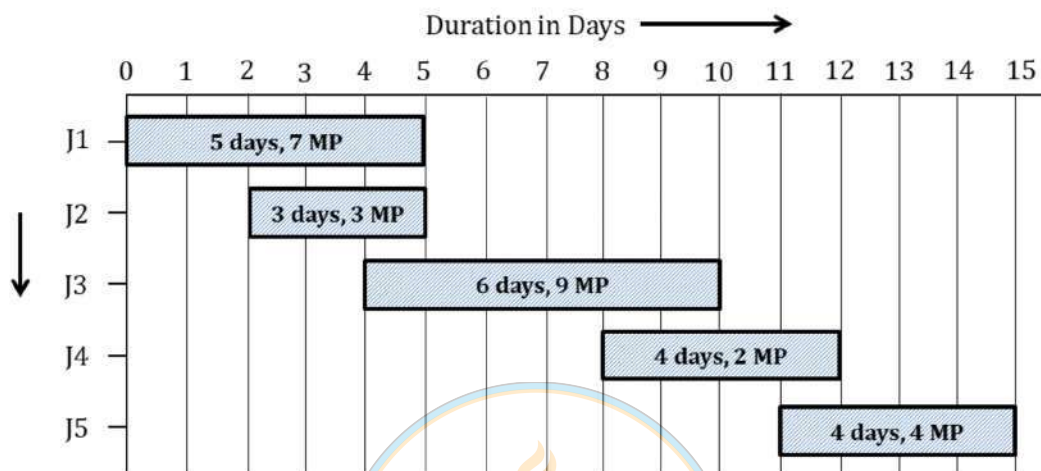
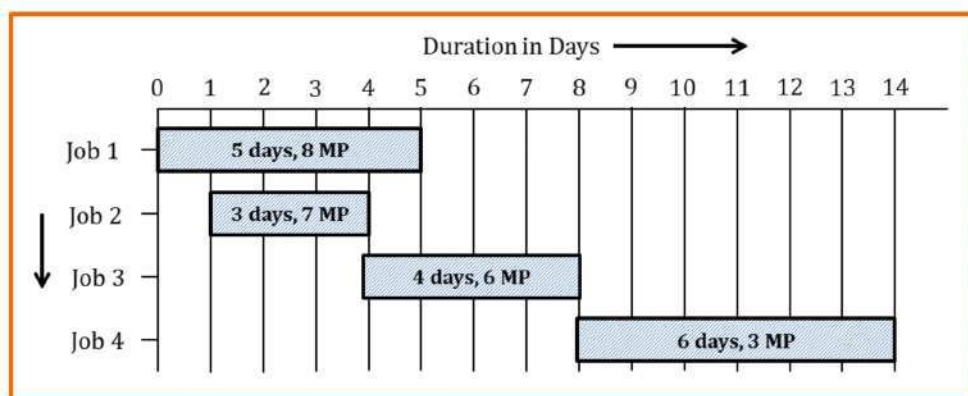


Fig. Gantt chart

Note: For J5: Start day = 11, Duration = 4 Therefore Total division of Duration = 11+4 = 15 Divisions

Q. Construct Gantt chart for the following. (Model QP)

Jobs	Start time	Duration	Manpower
1	0	5	8
2	1	3	7
3	4	4	6
4	8	6	3



8+6=14 divisions

Note:
MP = Man Power

5.5 Milestone Charts

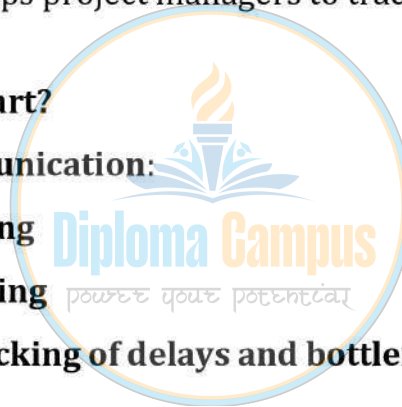
- ☐ A **Milestone Chart** is a project management tool used to track major events or achievements of a project.
- ☐ Milestone charts focus on key points that indicate progress or major stages.

Key Features of a Milestone Chart

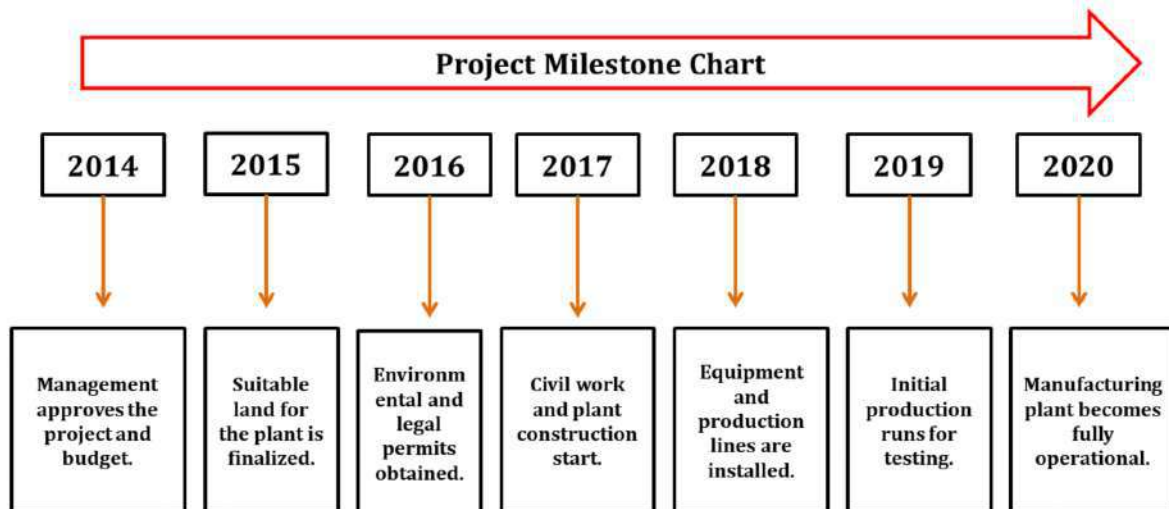
- Each milestone marks an important event, such as project initiation, phase completion or handing over the deliverables to the customers.
- The milestones are placed along a time scale to provide a visual sequence.
- Milestone charts do not show task durations—only key points.
- Milestone charts helps project managers to track progress.

Why Use a Milestone Chart?

- It improves Communication:
- It improves Planning
- It helps in Monitoring
- It helps in easy tracking of delays and bottlenecks.
- It helps in Reporting



Example of a Milestone Chart



5.6 Critical Path Method (CPM)

- ❑ The **Critical Path Method (CPM)** is a project management technique used to determine the **longest sequence of dependent tasks** that must be completed on time to complete the project as scheduled.
- ❑ It helps identify **critical tasks** that, if delayed, would delay the overall project.

Concepts of CPM

- **Activities (Tasks):** Work that must be completed.
- **Dependencies:** Some tasks must be completed before others can start.
- **Duration:** Estimated time required for each task.
- **Critical Path:** The longest sequence of tasks that determines the project's shortest completion time.

5.7 Construction of a Network

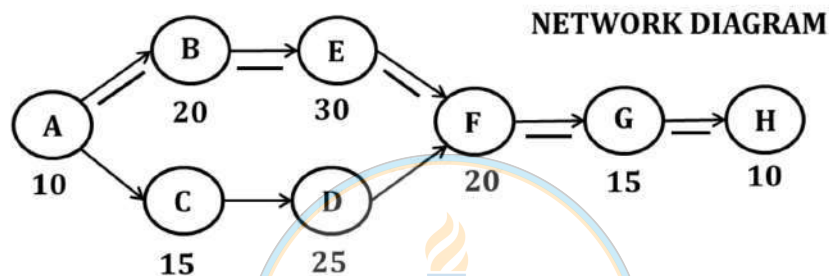
Steps in CPM

- i) **List all activities:** Identify project tasks and their durations.
- ii) **Determine dependencies:** Define which tasks must follow others.
- iii) **Draw a network diagram:** Represent tasks and dependencies visually.
- iv) **Estimate task durations:** Assign realistic timeframes.
- v) **Identify the longest path (Critical Path):** This determines the project's minimum completion time.
- vi) **Monitor and manage the path:** Ensure that critical tasks stay on schedule.

Example for CPM:

Activity	Predecessor(s)	Duration (Days)	Description
A	-	10	Site Selection
B	A	20	Obtain Permits
C	A	15	Land Preparation
D	C	25	Foundation Construction
E	B	30	Equipment Procurement
F	D, E	20	Equipment Installation
G	F	15	Trial Production
H	G	10	Full-Scale Production

Solution:



Path:

Path 01: A-B-E-F-G-H = 10+20+30+20+15+10 = 105 DAYS (CRITICAL PATH)

Path 02: A-C-D-F-G-H = 10+15+25+20+15+10 = 95 DAYS

Therefore path 01, A-B-E-F-G-H with duration of 105 days is the Critical Path and it is identified using double line in the network diagram.

5.8 Network Technique in Project Scheduling (PERT)

- ❑ **PERT (Program Evaluation and Review Technique)** is a project management technique used for scheduling, organizing and coordinating tasks.
- ❑ It is particularly useful for projects with **uncertain activity durations**, such as research, development or industrial projects.

Key Features of PERT

- It uses Activity-Based approach and breaks down a project into tasks and their dependencies.
- It uses three-time estimates (Optimistic, Most Likely, and Pessimistic) to find uncertainty.

- In the Network Diagram, circle represents events and arrow represents activity.
- Critical Path is identified by finding the longest sequence of dependent tasks.
- It helps in Risk Management.

Steps to Implement PERT

- Identify project tasks and define their dependencies.
- Estimate time for each task using:
 - Optimistic Time (O):** Best-case scenario.
 - Most Likely Time (M):** Normal conditions.
 - Pessimistic Time (P):** Worst-case scenario.
- Calculate Expected Time (TE) using the formula:

$$t_e = \frac{t_o + (4t_m) + t_p}{6}$$
- Create a PERT network diagram showing task dependencies.
- Determine the Critical Path by identifying the longest sequence of dependent tasks.
- Evaluate project completion time using statistical methods (variance and probability distribution).

Q. Calculate the expected time for each activity and also find the standard deviation: (Sep Makeup 2023)

Activity	t_o	t_m	t_p
1-2	9	12	21
1-3	6	12	18
2-4	1	1.5	5
3-4	4	8.5	10
4-5	10	14	24

Solution:

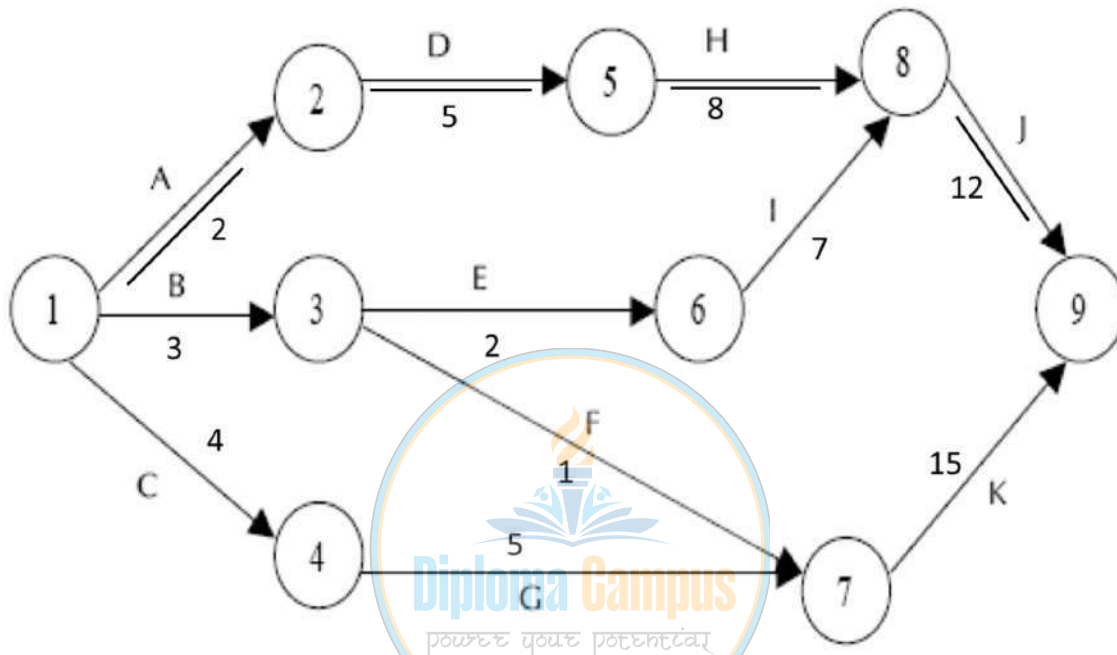
Activity	t_o	t_m	t_p	Expected Time , $t_e = \frac{t_o + 4t_m + t_p}{6}$	Variance $\sigma^2 = \left[\frac{t_o - t_p}{6} \right]^2$
1-2	9	12	21	13	4
1-3	6	12	18	12	4
2-4	1	1.5	5	2	0.444
3-4	4	8.5	10	8	1
4-5	10	14	24	15	5.444

Standard Deviation, $\sigma = \sqrt{(4 + 4 + 0.444 + 1 + 5.444)} = \underline{\underline{3.858}}$

Q. Construct a network diagram and find the critical path for a project.

ACTIVITY	A	B	C	D	E	F	G	H	I	J	K
PREDECESSOR	-	-	-	A	B	B	C	D	E	H,I	F,G
DURATION IN DAYS	2	3	4	5	2	1	5	8	7	12	15

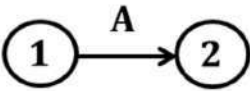
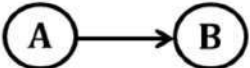
Solution:



Path	Duration (days)
1-2-5-8-9	2+5+8+12=27 (critical Path)
1-3-6-8-9	3+2+7+12=24
1-3-7-9	3+1+15=19
1-4-7-9	4+5+15=24

The Path: **1-2-5-8-9**, with duration of 27days is the critical path and it is identified using double lines in Network Diagram.

Difference between PERT and CPM.

Sl. No	Index	PERT	CPM
1.	Full Form	Project Evaluation and Review Technique	Critical Path Method
2.	Focus on	Time estimation & risk	Cost, time & resource
3.	Time calculation	Uses three time estimates (t_o , t_m & t_p).	Uses fixed time estimates
4.	Best For	Research & Development projects (large scale projects).	Construction projects (complex projects).
5.	Nature	Probabilistic	Deterministic
6.	Type of Activities	Non – repetitive activities	Repetitive activities
7.	Orientation	Event-oriented.	Activity-oriented.
8.	Representation	Circle for events & line connecting the circles represents an activity. 	Circle for an activity and the line joining the circles represents an events. 
9.	Flexibility	More flexible and suitable for uncertain environments.	Less flexible but highly efficient for well-defined projects.
10.	Example	Finding a new vaccine.	Construction of dams, bridges or roads.

5.9 Project Review

- ☐ A **Project Review** is a planned valuation of a project progress, performance and results.
- ☐ It helps in identifying strength, weakness, risk and areas for improvement.
- ☐ There are different types of project reviews depending on the project stage and purpose.

Types of Project Reviews

- Progress Review
- Technical Review
- Financial Review
- Risk Review
- Final Review (Post-Project Review)

Objectives of project review:

- To examine whether the project is completed as per the plan or not.
- To find whether the project is completed on given time or not.
- To find whether the project is completed on given cost or not.
- To find whether the resources are sufficient or not.
- To measure the impact of the project.
- To examine the project efficiency.
- To measure the quality of the project.
- To review the safety features followed during the project.
- To examine the methods, process, procedures followed during the project.
- To measure the outcome of the project.
- Lesson learned after the project completion to improve future projects.

5.10 Initial Review and its types

- ☐ An **initial review** is a first look at project before conducting detailed examination.
- ☐ It helps to correct or meets certain standards.
- ☐ It helps in finding problems early and making improvements.

Types of Initial Review

- i) **Document Review:** Checking papers, reports, mistakes and missing details.
- ii) **Project Review:** Checking project plan possible or not.

- iii) **Financial Review:** Checking budgets and expenses.
- iv) **Performance Review:** Checking, how well a person or team is.
- v) **Product Review:** Testing a product before selling.
- vi) **Legal Review:** Checking contracts and agreements.
- vii) **Risk Review:** Identifying possible risks in a project.
- viii) **Compliance Review:** Making sure rules and regulations are being followed.

5.11 Objectives of Project Audit

- To evaluate project progress.
- To manage project risk.
- To manage resources.
- To manage quality
- To manage standards.
- To manage budget (cost).
- To keep proper records.
- To satisfy stakeholders.
- To improve communication.
- To improve project efficiency.

5.12 Functions of Project Auditor

- To make sure project follows rules, regulations and standards.
- To make sure project is completed as per the plan, objective, time and cost.
- To identify the potential risk.
- To check the project budget and expenses are accurate.
- To ensure time, money and men are properly utilized.
- To ensure proper quality.
- To ensure effective communication.
- To provide suggestion improvements for better project management.

PROBLEMS FROM OLD QUESTION PAPERS:

1] Determine the expected time for the following activities: (Model QP-01)

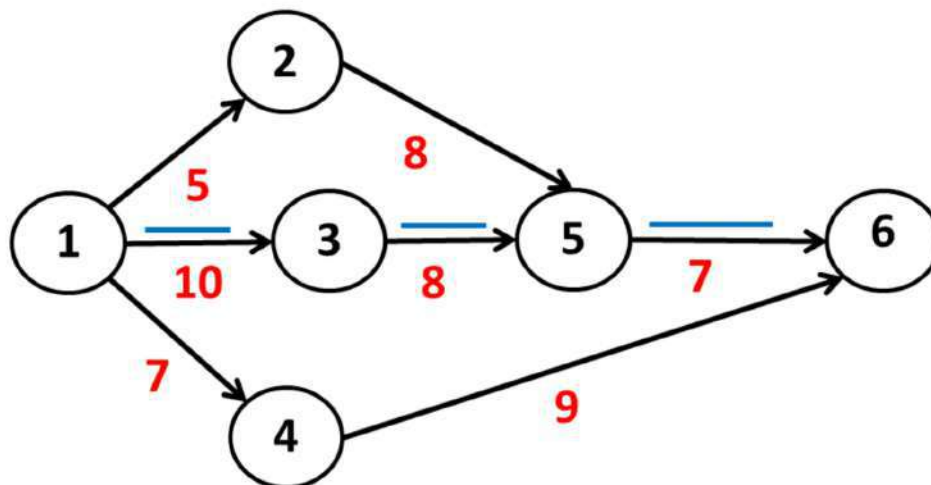
Activity	Estimated duration in days		
	Optimistic	Most Likely	Pessimistic
1 - 2	2	5	8
1 - 3	4	10	16
1 - 4	1	7	13
2 - 5	5	8	11
3 - 5	2	8	14
4 - 6	6	9	12
5 - 6	4	7	10

Solution:

Activity	Estimated duration in days			Expected Time $t_e = \frac{t_o + 4t_m + t_p}{6}$
	Optimistic (to)	Most Likely (tm)	Pessimistic (tp)	
1 - 2	2	5	8	5
1 - 3	4	10	16	10
1 - 4	1	7	13	7
2 - 5	5	8	11	8
3 - 5	2	8	14	8
4 - 6	6	9	12	9
5 - 6	4	7	10	7

Extra:

To construct Network Diagram for the above activities, Expected time must be considered to draw network diagram:



Network Diagram

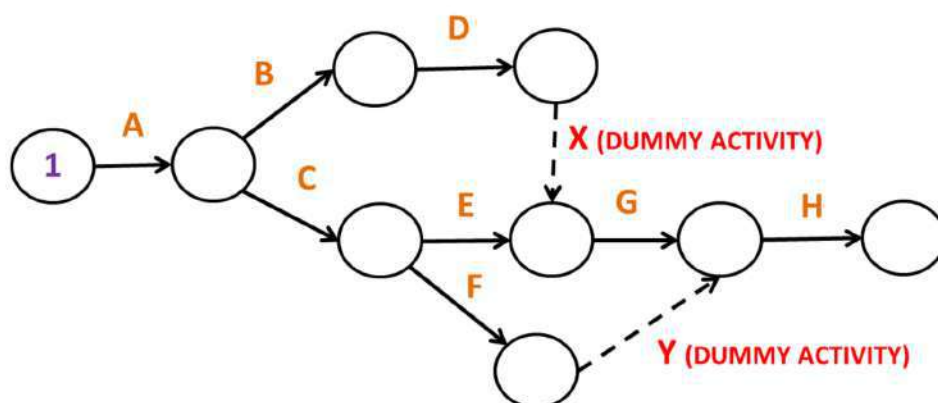
Path No.	Path	Sum of duration	Total Days	Remarks
1	1-2-5-6	5+8+7	=20 Days	
2	1-3-5-6	10+8+7	=25 Days	Critical Path
3	1-4-6	7+9	=16 Days	

Therefore, path 2 with duration of 25 days is the Critical Path and it is identified using double lines in the network diagram.

2] Develop a network diagram for a project whose activities and preceding activities with duration are given below: (Model QP 01, May 2024)

Activity	Predecessor	Duration (Days)
A	-	3
B	A	4
C	A	2
D	B	5
E	C	1
F	C	2
G	D,E	4
H	F,G	3

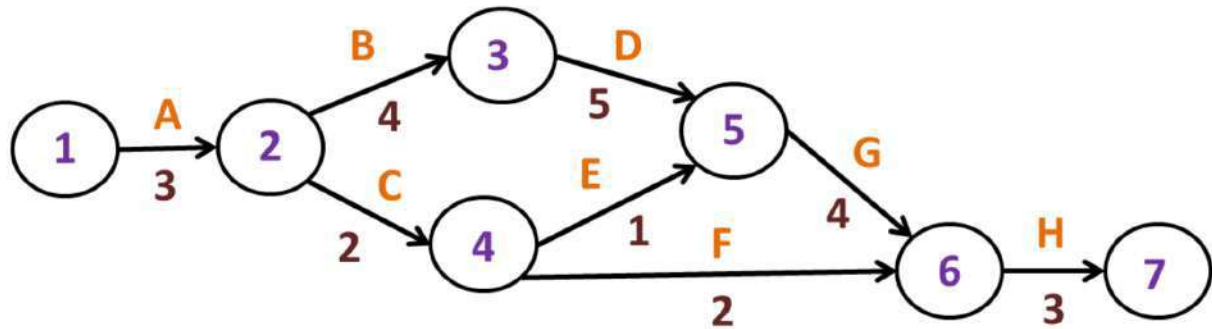
Solution 01 – Using PERT Method:



Rough Network Diagram

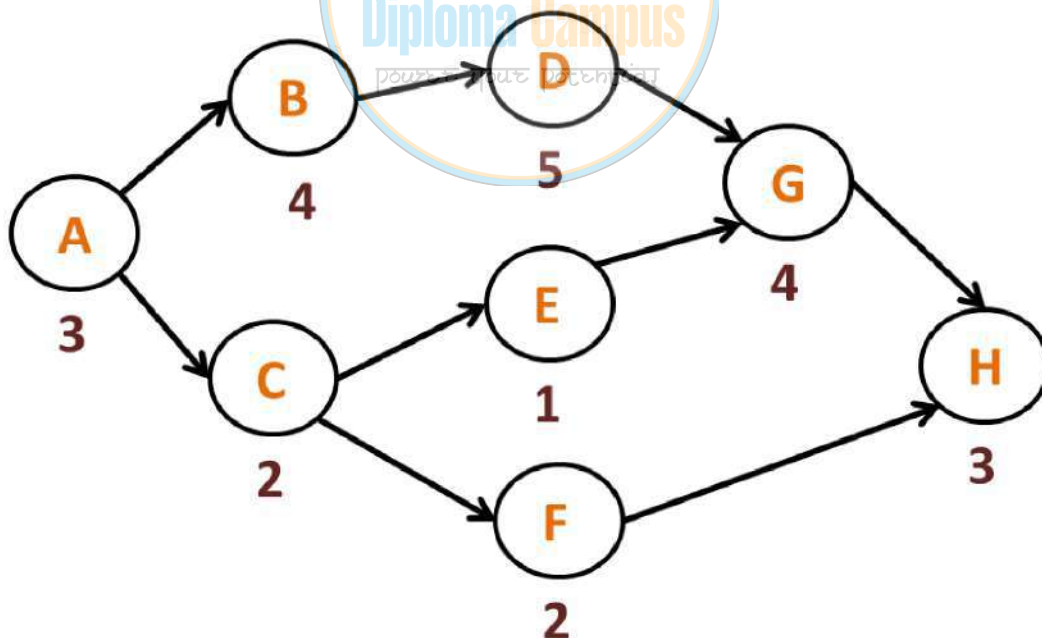
Note: D, E to G and F, G to H is shown using Dummy activity "X" and "Y" using dashed lines.

Now actual Network Diagram is drawn by taking out one of the circles from dummy activity.



PERT - Network Diagram

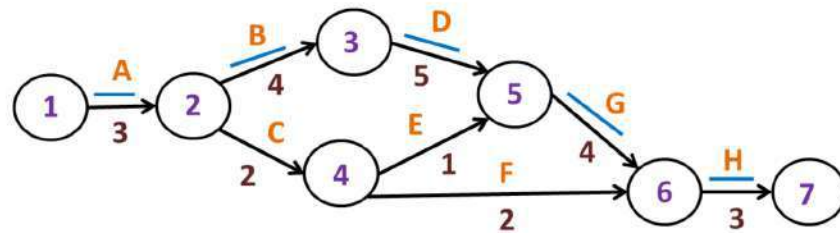
Solution 02 – Using CPM Method:



CPM - Network Diagram

Extra:

Solution 01 – Using PERT Method – Finding the Critical Path:

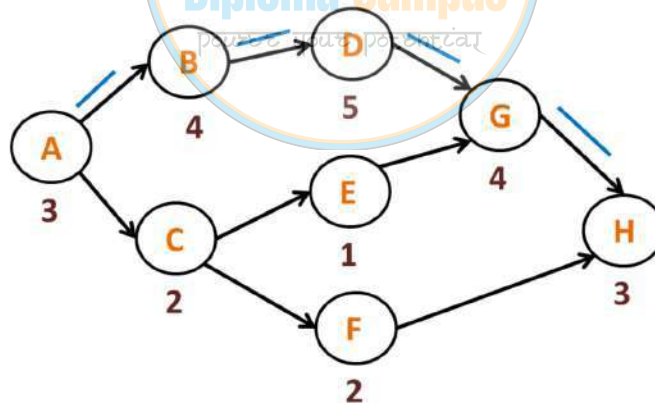


Path No.	Path	Sum of duration	Total Days	Remarks
1	1-2-3-5-6-7	3+4+5+4+3	=19 Days	Critical Path
2	1-2-4-5-6-7	3+2+1+4+3	=13 Days	
3	1-2-4-6-7	3+2+2+3	=10 Days	

Therefore, path 1 with duration of 19 days is the Critical Path and it is identified using double lines in the network diagram.

OR

Solution 02 – Using CPM Method – Finding the Critical Path:



Path No.	Path	Sum of duration	Total Days	Remarks
1	A-B-D-G-H	3+4+5+4+3	=19 Days	Critical Path
2	A-C-E-G-H	3+2+1+4+3	=13 Days	
3	A-C-F-H	3+2+2+3	=10 Days	

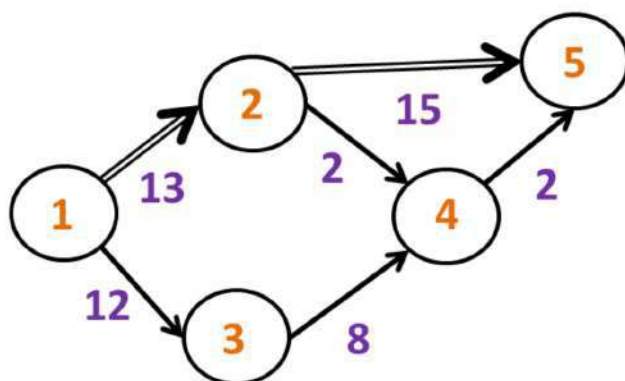
Therefore, path 1 with duration of 19 days is the Critical Path and it is identified using double lines in the network diagram.

3] The time estimate (t_o), (t_m) and (t_p) for each activity in a project are shown below. Determine the expected time for each activity, draw the PERT network, critical path and total time required to complete the project. (JULY MAKEUP 2024, Model QP 02)

Activity	Time estimate		
	t_o	t_m	t_p
1-2	9	12	21
1-3	6	12	18
2-4	1	1.5	5
3-4	4	8.5	10
2-5	10	14	24
4-5	1	2	3

Solution:

Activity	Time estimate			Expected Time $t_e = \frac{t_o + 4t_m + t_p}{6}$
	t_o	t_m	t_p	
1-2	9	12	21	13
1-3	6	12	18	12
2-4	1	1.5	5	2
3-4	4	8.5	10	8
2-5	10	14	24	15
4-5	1	2	3	2



Network Diagram

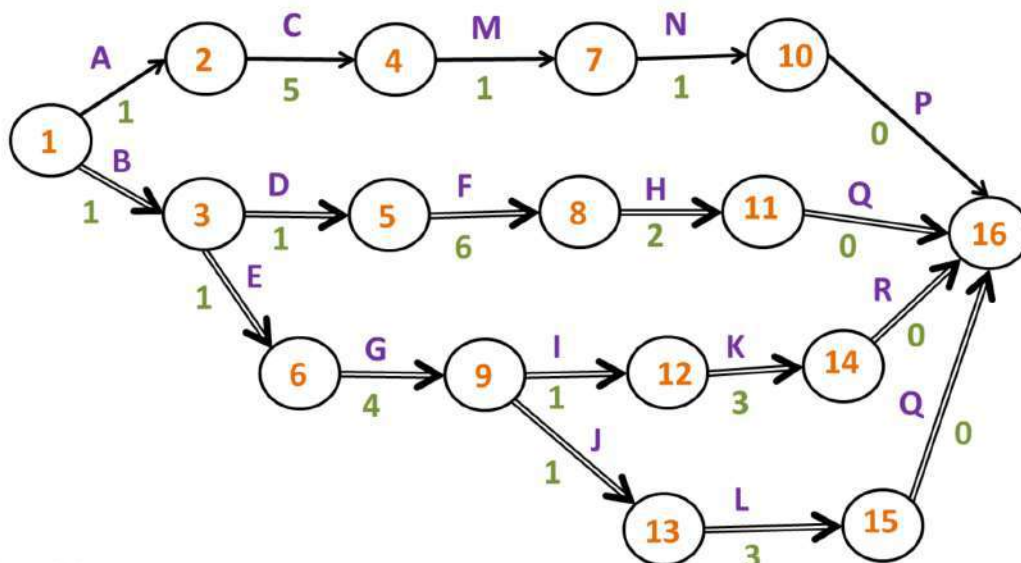
Here Critical Path is identified Using double lines.

Path No.	Path	Sum of duration	Total Days	Remarks
1	1-2-5	13+15	=28 Days	Critical Path
2	1-2-4-5	13+2+2	=17 Days	
3	1-3-4-5	12+8+2	=22 Days	

4] Construct the Network diagram and find the Critical path for the following data (Model QP 02).

Task	Predecessor(s)	Duration (days)
A	None	1
B	None	1
C	A	5
D	B	1
E	B	1
F	D	6
G	E	4
H	F	2
I	G	1
J	G	1
K	I	3
L	J	3
M	C	1
N	M	1

Solution 01: Network Diagram using PERT Method:



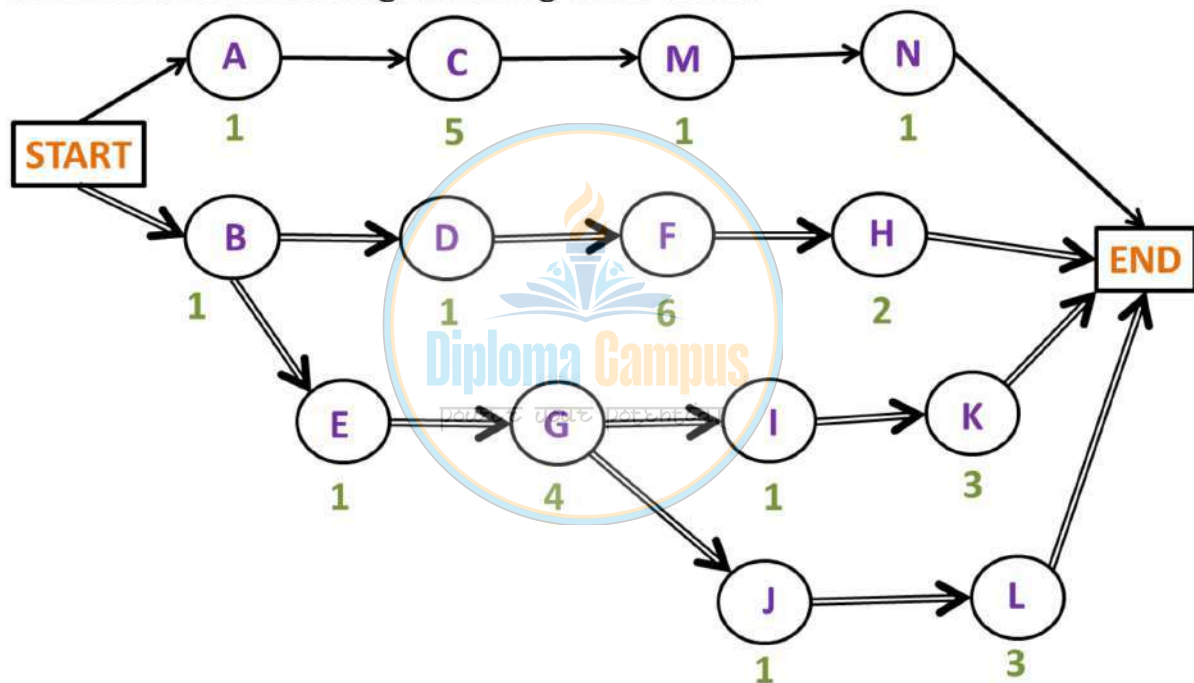
Network Diagram _ PERT Method

Here the critical path is identified using double lines.

Path No.	Path	Sum of duration	Total Days	Remarks
1	1-2-4-7-10-16	1+5+1+1+0	=8 Days	
2	1-3-5-8-11-16	1+1+6+2+0	=10 Days	Critical Path
3	1-3-6-9-12-14-16	1+1+4+1+3+0	=10 Days	Critical Path
4	1-3-6-9-13-15-16	1+1+4+1+3+0	=10 Days	Critical Path

OR

Solution 02: Network Diagram using CPM Method



Network Diagram _ CPM Method

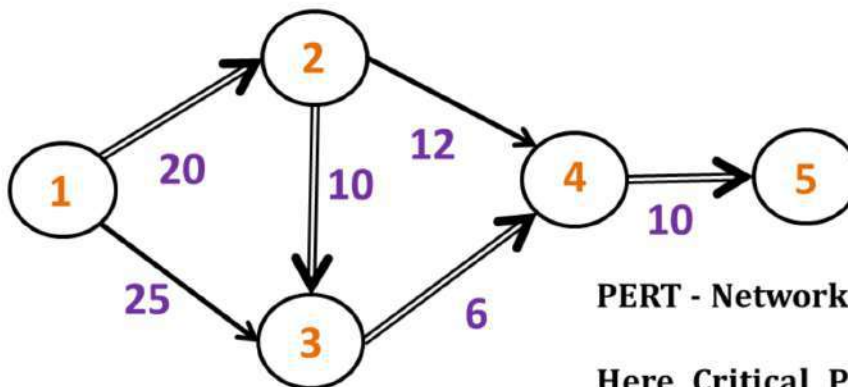
Here the critical path is identified using double lines.

Path No.	Path	Sum of duration	Total Days	Remarks
1	A-C-M-N	1+5+1+1	=8 Days	
2	B-D-F-H	1+1+6+2	=10 Days	Critical Path
3	B-E-G-I-K	1+1+4+1+3	=10 Days	Critical Path
4	B-E-G-J-L	1+1+4+1+3	=10 Days	Critical Path

5] Construct the Network diagram and identify the Critical path.
(NOV-DEC 2024)

Activity	1-2	1-3	2-3	2-4	3-4	4-5
Duration (in Days)	20	25	10	12	6	10

Solution:



PERT - Network Diagram

Here Critical Path is identified
Using double lines.

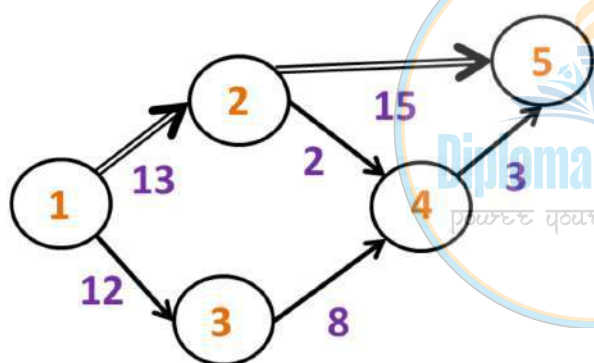
Path No.	Path	Sum of duration	Total Days	Remarks
1	1-2-4-5	20+12+10	=42 Days	
2	1-2-3-4-5	20+10+6+10	=46 Days	Critical Path
3	1-3-4-5	25+6+10	=41 Days	

6] Develop the network diagram for the following activities with the corresponding time estimates. (May 2024)

Activity	Time estimate		
	To	Tm	Tp
1-2	8	12	21
1-3	6	12	18
2-4	1	1.5	3
3-4	4	8	9
2-5	10	14	23
4-5	2	2	3

Solution:

Activity	Time estimate			Expected Time $t_e = \frac{t_o + 4t_m + t_p}{6}$	Expected Time $t_e = \frac{t_o + 4t_m + t_p}{6}$ (Round up to next value)
	t_o	t_m	t_p		
1-2	8	12	21	12.8	13
1-3	6	12	18	12.0	12
2-4	1	1.5	3	1.7	2
3-4	4	8	9	7.5	8
2-5	10	14	23	14.8	15
4-5	2	2	3	2.2	3



PERT - Network Diagram

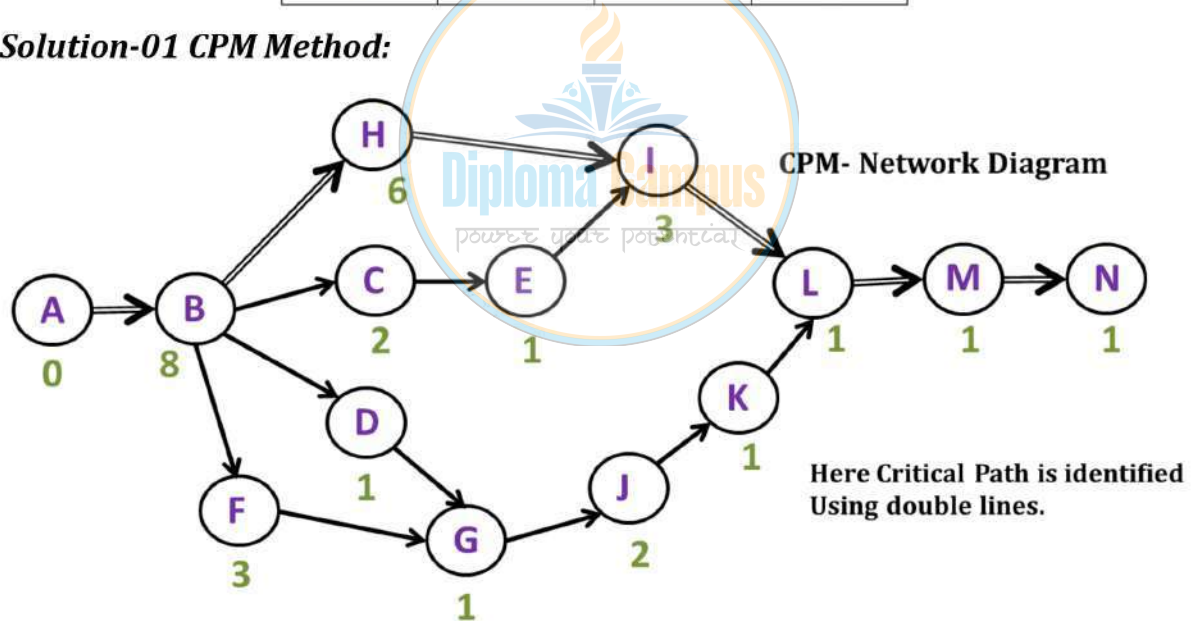
Here Critical Path is identified Using double lines.

Path No.	Path	Sum of duration	Total Days	Remarks
1	1-2-5	13+15	=28 Days	Critical Path
2	1-2-4-5	13+2+2	=17 Days	
3	1-3-4-5	12+8+3	=23 Days	

7] Construct a Network diagram for the following shown activities. Determine the critical path and total time required to complete the project. (June July 2023, July Makeup 2024)

Activity Letter	Preceding Activities	Duration (days)	No. of Employees needed
A	Nil	0	---
B	A	8	3
C	B	2	5
D	B	1	2
E	C	1	3
F	B	3	7
G	D,F	1	2
H	B	6	5
I	E,H	3	3
J	G	2	4
K	J	1	3
L	I,K	1	3
M	L	1	2
N	M	1	2

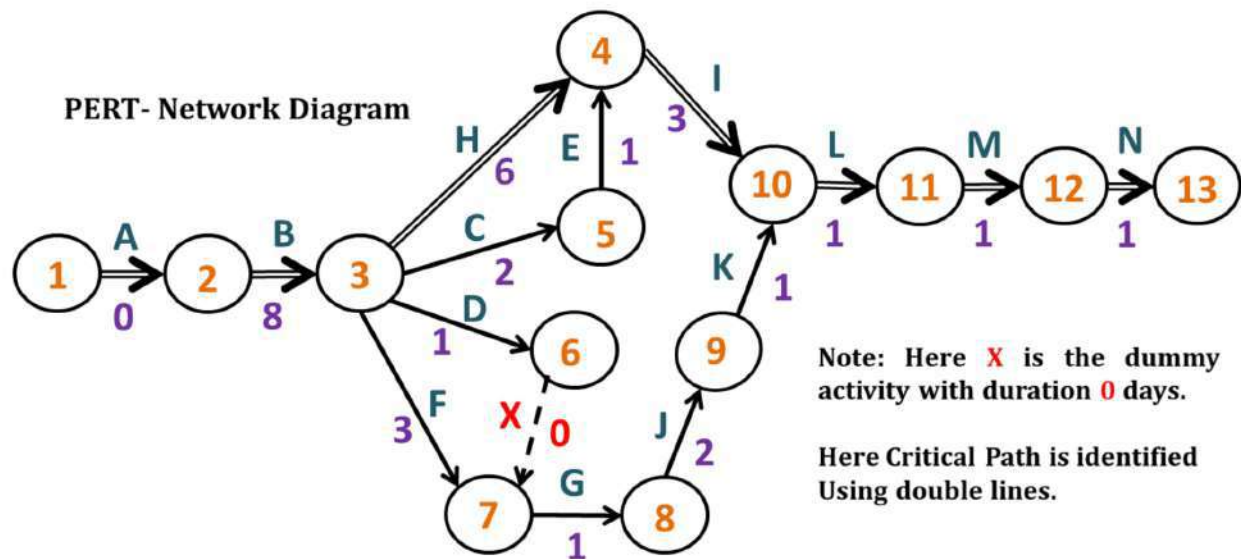
Solution-01 CPM Method:



Path No.	Path	Sum of duration	Total Days	Remarks
1	A-B-H-I-L-M-N	0+8+6+3+1+1+1	=20 Days	Critical Path
2	A-B-C-E-I-L-M-N	0+8+2+1+3+1+1+1	=17 Days	
3	A-B-D-G-J-K-L-M-N	0+8+1+0+1+2+1+1+1+1	=16 Days	
4	A-B-F-G-J-K-L-M-N	0+8+3+1+2+1+1+1+1	=18 Days	

OR

Solution-02 PERT Method:



Path No.	Path	Sum of duration	Total Days	Remarks
1	1-2-3-4-10-11-12-13	0+8+6+3+1+1+1	=20 Days	Critical Path
2	1-2-3-5-4-10-11-12-13	0+8+2+1+3+1+1+1	=17 Days	
3	1-2-3-6-7-8-9-10-11-12-13	0+8+1+0+1+2+1+1+1+1	=16 Days	
4	1-2-3-7-8-9-10-11-12-13	0+8+3+1+2+1+1+1+1	=18 Days	

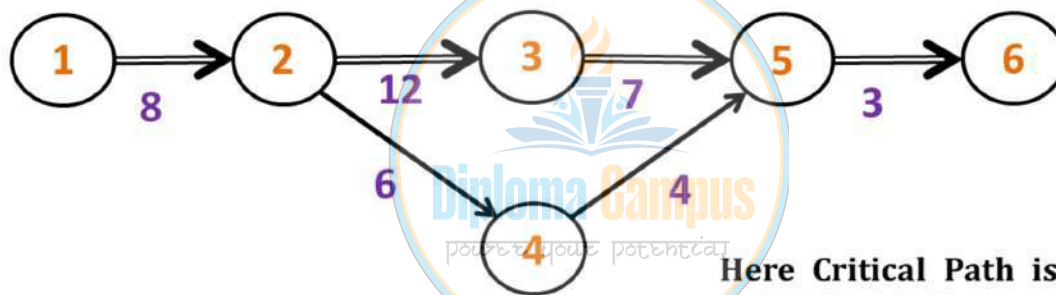
9] The following activities involved as shown in the table below, estimate time for each activity. Draw the PERT Network, determine total time required for completion of project and find the critical path. (Dec 2023)

Activities	T_o	T_m	T_p
1-2	7	8	9
2-3	11	12	13
2-4	5	6	7
3-5	6	7	8
4-5	3	4	5
5-6	2	3	4

Solution:

Activities	t_o	t_m	t_p	Expected Time $t_e = \frac{t_o + 4t_m + t_p}{6}$
1-2	7	8	9	8
2-3	11	12	13	12
2-4	5	6	7	6
3-5	6	7	8	7
4-5	3	4	5	4
5-6	2	3	4	3

PERT - Network Diagram



Here Critical Path is identified Using double lines.

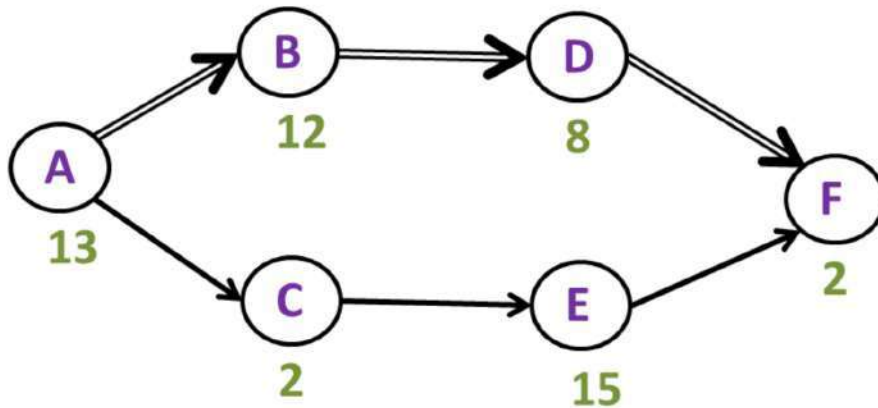
Path No.	Path	Sum of duration	Total Days	Remarks
1	1-2-3-5-6	8+12+7+3	=30 Days	Critical Path
2	1-2-4-5-6	8+6+4+3	=21 Days	

10] Construct the network diagram whose activities and their predecessor relationship is given in below table. Determine the total time required for completion of project and find the critical path. (Dec 2023)

Activity	A	B	C	D	E	F
Predecessor	--	A	A	B	C	D,E
Time	13	12	2	8	15	2

Solution 01 _ using CPM method:

CPM – Network Diagram



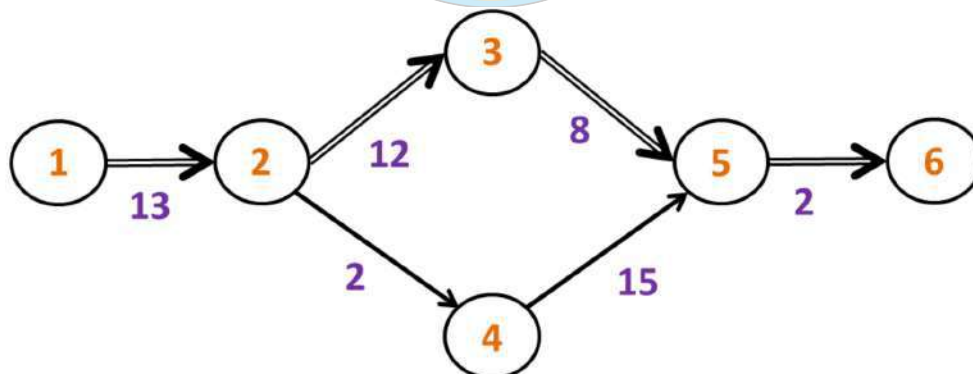
Here Critical Path is identified using double lines.

Path No.	Path	Sum of duration	Total Days	Remarks
1	A-B-D-F	13+12+8+2	=35 Days	Critical Path
2	A-C-E-F	13+2+15+2	=32 Days	

OR

Solution 02 _ using PERT method:

PERT - Network Diagram



Here Critical Path is identified using double lines.

Path No.	Path	Sum of duration	Total Days	Remarks
1	1-2-3-5-6	13+12+8+2	=35 Days	Critical Path
2	1-2-4-5-6	13+2+15+2	=32 Days	

11] The three time estimate (to), (tm), (tp) for each activity in a project are given below. Determine the expected time for each activity and also calculate the Standard Deviation of the project. (Feb March 2023)

Activity	t _o	t _m	t _p
1-2	9	12	21
1-3	6	12	18
2-4	1	1.5	5
3-4	4	8.5	10
2-5	10	14	24

Solution:

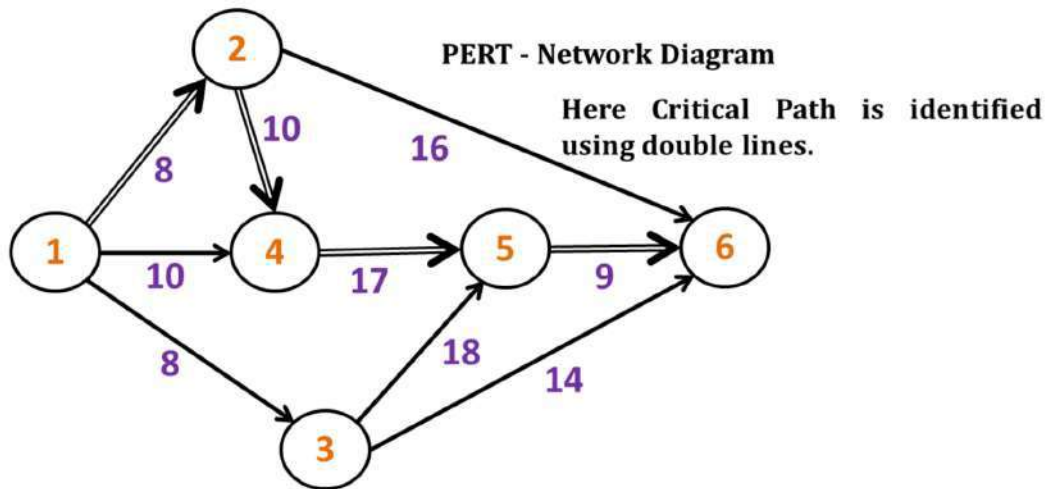
Activity	t _o	t _m	t _p	Expected Time , $t_e = \frac{t_o + 4t_m + t_p}{6}$	Variance $\sigma^2 = \left[\frac{t_o - t_p}{6} \right]^2$
1-2	9	12	21	13	4
1-3	6	12	18	12	4
2-4	1	1.5	5	2	0.444
3-4	4	8.5	10	8	1
2-5	10	14	24	15	5.444

Standard Deviation, $\sigma = \sqrt{(4 + 4 + 0.444 + 1 + 5.444)} = \underline{\underline{3.858}}$

12] The following activities constitute a project (i) construct the network to represent the project and also identify the critical path (ii) write all project paths possible. (Feb March 2023)

Activity	Duration (Days)
1-2	8
1-3	8
1-4	10
2-4	10
2-6	16
3-5	18
3-6	14
4-5	17
5-6	9

Solution:



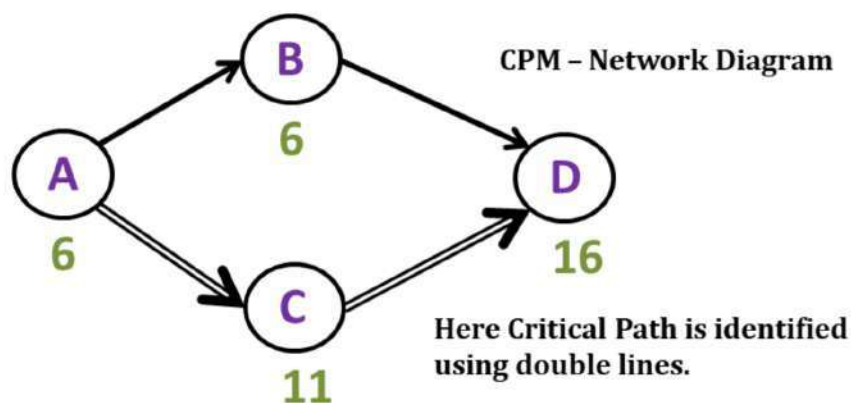
The various paths of the network and their duration are:

Path No.	Path	Sum of duration	Total Days	Remarks
1	1-2-6	8+16	=24 Days	
2	1-2-4-5-6	8+10+17+9	=44 Days	Critical Path
3	1-4-5-6	10+17+9	=36 Days	
4	1-3-5-6	8+18+9	=35 Days	
5	1-3-6	8+14	=22 Days	

13] Prepare an early start schedule network diagram for the given project. (Nov Dec 2022)

Activity	Predecessor	Duration
A	---	6
B	A	6
C	A	11
D	B,C	16

Solution:

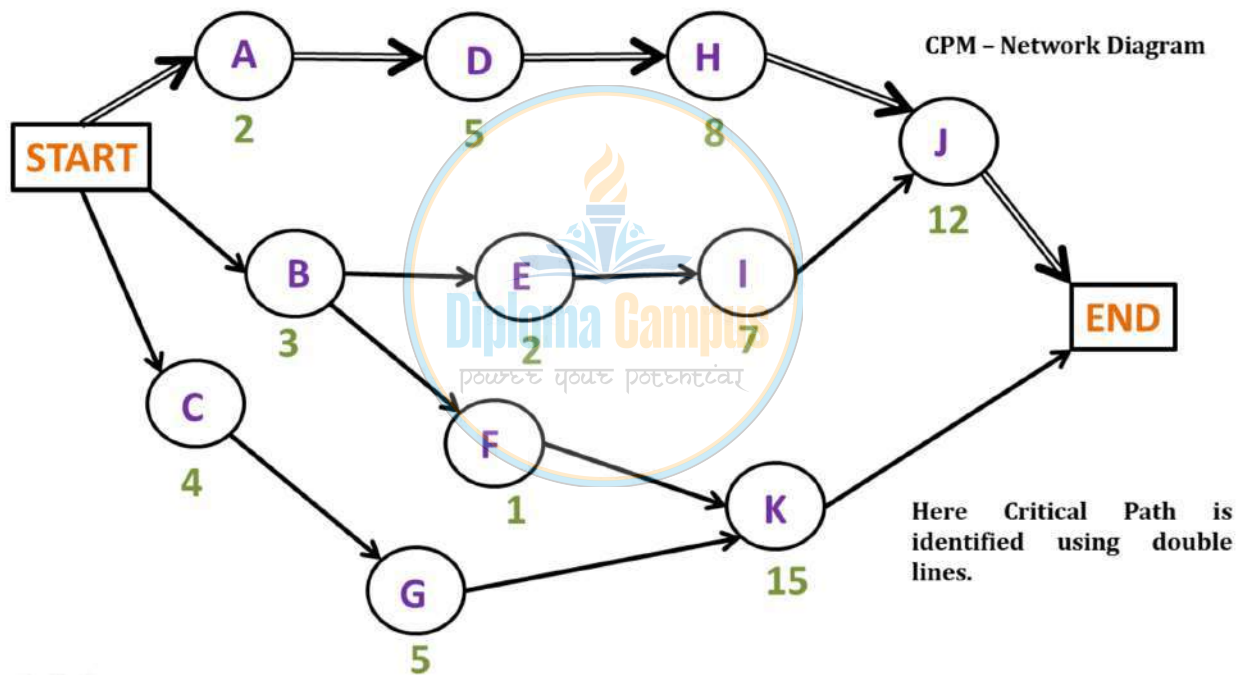


Path No.	Path	Sum of duration	Total Days	Remarks
1	A-B-D	6+6+16	=28 Days	
2	A-C-D	6+11+16	=33 Days	Critical Path

14] Construct a network diagram and find the critical path for a project.

ACTIVITY	A	B	C	D	E	F	G	H	I	J	K
PREDECESSOR	-	-	-	A	B	B	C	D	E	H,I	F,G
DURATION IN DAYS	2	3	4	5	2	1	5	8	7	12	15

Solution:



Path No.	Path	Sum of duration	Total Days	Remarks
1	A-D-H-J	2+5+8+12	=27 Days	Critical Path
2	B-E-I-J	3+2+7+12	=24 Days	
3	B-F-K	3+1+15	=19 Days	
4	C-G-K	4+5+15	=24 Days	

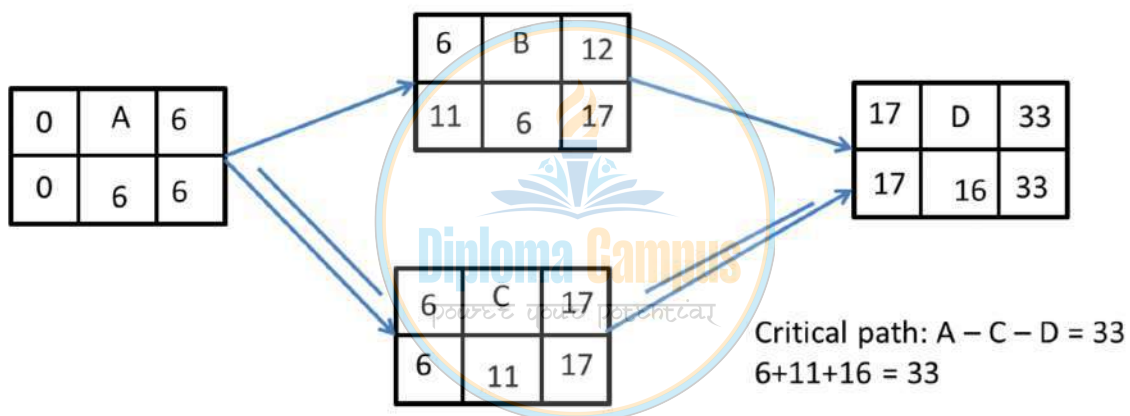
**15] Prepare an early start schedule network diagram for the given project.
(Nov-Dec-2022-Makeup Exam)**

Activity	Predecessor	Duration
A	---	6
B	A	6
C	A	11
D	B,C	16

Solution:

ACTIVITY	PREDECESSOR	DURATION
A	---	6
B	A	6
C	A	11
D	B, C	16

Early start	Activity	Early Finish
Late start	Duration	Late Finish



Here the Critical Path is identified using double lines.

5.13 Case Studies

Case Study 4: A company is started a new project with several activities.

The project involves 5 activities.

Activity Identification and Sequencing:

- Activity A: (Duration: 5 days)
- Activity B: (Duration: 10 days, follows Activity A)
- Activity C: (Duration: 15 days, follows Activity B)
- Activity D: (Duration: 12 days, follows Activity C)
- Activity E: (Duration: 8 days, follows Activity D)

Dependencies:

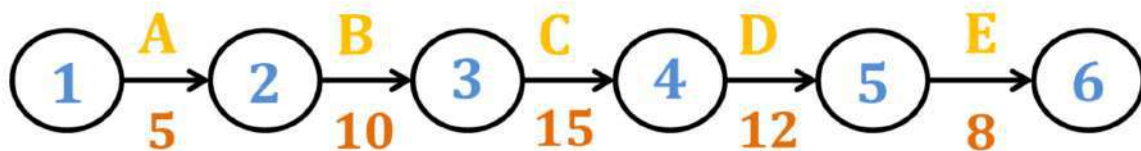
- Activity A must precede Activity B (Finish to Start dependency).
- Activity B must precede Activity C.
- Activity C must precede Activity D.
- Activity D must precede Activity E.

Construct a network diagram, identify the critical path and determine the Total project duration.

Case Study 4 Solution:

Activity	Predecessor	Duration (Days)
A	---	5
B	A	10
C	B	15
D	C	12
E	D	8

Network Diagram using PERT method:

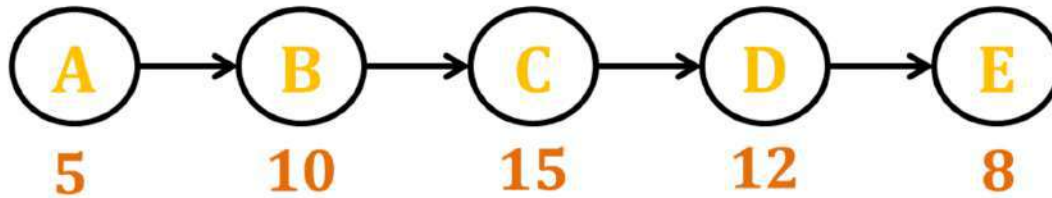


Critical Path: 1-2-3-4-5-6 = 5+10+15+12+8 = 50 Days

∴ Total Project Duration is 50 days

OR

Network Diagram using CPM method:



Critical Path: A-B-C-D-E = 5+10+15+12+8 = 50 Days

∴ Total Project Duration is 50 days



